
IAL Chemistry Unit 3

Rapid Revision

WCH13 — Practical Skills in Chemistry I

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* guide (21 past papers, Jun 2019 - Jun 2026)

50 marks · 1 h 20 min · 4–5 structured questions

No lab work: you describe methods, give observations, process data and evaluate. Most marks are for exact mark-scheme wording and clean calculation chains.

TRIAGE — WHAT APPEARS, AND HOW OFTEN

Topic	of 21	Marks
Titration: colours, results table, mean, multi-step calculation	~17	8-19
Organic preparation: reflux, distillation, separating funnel, drying, % yield	~15	8-18
Flame test method and colours	~14	2-5
Enthalpy: calorimetry, extrapolation, errors	~13	8-13
Inorganic ion tests (halide, sulfate, carbonate, NH_4^+ , nitrate)	~12	4-11
Organic functional-group tests	~12	2-8
Graphs: plotting, best fit, extrapolation, tangents	~12	2-5
Uncertainty, % error, improvements	~10	1-4
Gas volumes, molar volume, identifying a metal	~7	3-4
Hazard symbols and precautions	most	1-2

THE MARK-SCHEME RULES THAT DECIDE A* vs A

Ions, not elements: Ba^{2+} not Ba · **iodide** / I^- not iodine · "**bromide ion**" not bromine ion. **Halogen** $\neq$ **halide**.

State the state: "**white precipitate**", never "white" or "white solution". **Steamy fumes** $\neq$ **white smoke**. "Clear" is ignored — write **colourless**.

Contradictions cancel: an extra wrong observation loses the mark. The **list principle** applies.

Test + result: the result mark **depends on a correct test**. Always give both.

Name and formula: if you give both, **both must be right**. Give one unless asked for both.

Calculations: show every step (**transferred errors are credited**); don't round intermediates to 1 s.f.; final answer 2-3 s.f., **never 1 s.f.**

Enthalpy: sign + value + **kJ mol^{-1}** (" kJ/mol^{-1} " is penalised).

Vague words score zero: "more accurate" · "to ensure accuracy" · "impurities" · "to prevent bumping" (without why) · "safety" · "wear goggles/lab coat" for a specific hazard.

SHEET 1 — FLAME TESTS & INORGANIC TESTS

FLAME TEST METHOD — 4-5 MARKS, ASKED ALMOST EVERY YEAR

1. Use a **nichrome or platinum wire** (or silica rod) — **not "nickel" alone, not an inoculating loop**. Reason: **inert / gives no flame colour / high melting temperature**.
 2. Clean the wire by dipping in **concentrated hydrochloric acid** — only HCl, because **metal chlorides are more volatile**.
 3. Dip the moistened wire in the solid and hold it in a **hot, non-luminous (blue, roaring, air-hole open)** Bunsen flame. **Never "burn the sample"**.
 4. Observe the colour.
- Improvements:** HCl not HNO₃ (chlorides more volatile) · nichrome not copper (copper gives its own blue-green) · air-hole open (a luminous yellow flame masks the colour) · a wooden splint is cheap but **wood burns yellow**, masking the colour.

Ion	Write	Avoid
Li ⁺	red (carmine)	brick-red, orange; " crimson " was rejected in 2026
Na ⁺	yellow (orange / gold allowed)	orange-red
K ⁺	lilac	purple, pink, violet
Rb ⁺	red-violet / red-purple	brick-red, orange-red
Ca ²⁺	orange-red / brick-red	"red" or "orange" alone
Sr ²⁺	red / crimson (scarlet allowed)	orange
Ba ²⁺	(apple) green	blue-green
Mg ²⁺ , Be ²⁺	no colour	white

Ion	Test	Positive result and the key "why"
CO ₃ ²⁻	Dilute acid; bubble the gas into limewater	Effervescence; limewater cloudy/milky
SO ₄ ²⁻	Dilute HCl (or HNO ₃), then BaCl₂(aq)	White precipitate . The acid removes CO ₃ ²⁻ /SO ₃ ²⁻ , which also give white ppts. Never H₂SO₄ — it adds sulfate
Cl ⁻	Dilute HNO₃ , then AgNO ₃ (aq); then ammonia. Not HCl — it adds Cl ⁻	White ppt, dissolves in dilute NH₃
Br ⁻		Cream ppt, dissolves only in concentrated NH₃
I ⁻		Yellow ppt, insoluble in conc. NH₃
NH ₄ ⁺	NaOH(aq), warm; damp red litmus at the mouth , not dipped in	Gas turns it blue ; white smoke with HCl on a glass rod

Halide mixture: add **dilute** NH₃ — only AgCl dissolves; then **conc.** NH₃ — AgBr dissolves, AgI remains. A white + yellow mixture becomes "**more yellow**" after dilute NH₃.

Solid + conc. H₂SO₄: NaCl → **steamy fumes only** · NaBr → steamy fumes + **brown/red-brown fumes** (Br₂, SO₂) · NaI → steamy fumes, **purple vapour, black/grey solid, yellow solid, bad-egg smell** (I₂, S, H₂S). **Fume cupboard** — Br₂, SO₂, H₂S, HBr, HI are toxic.

Halogen displacement: add Cl₂(aq) or Br₂(aq), then hexane, shake. Upper organic layer **purple → I₂; orange → Br₂**. Bromine water + NaCl: **no reaction** (only paler from dilution).

Nitrate decomposition: Group 2 and Li nitrates give **brown NO₂ + O₂** (relights a glowing splint); Na/K nitrates give **nitrite + O₂ only**. Complete when **no more brown gas** / heated to constant mass.

Gas tests: H₂ squeaky pop · O₂ relights a glowing splint · CO₂ limewater milky · NH₃ damp red litmus blue · HCl steamy fumes, damp blue litmus red · H₂O anhydrous CuSO₄ **white → blue**, or cobalt chloride paper **blue → pink (not red)**.

SHEET 2 — ORGANIC TESTS

Group	Reagent	Positive result (exact)	Note
C=C alkene	Bromine water	orange/yellow/brown → colourless	Not "red" . Or acidified KMnO_4 , purple → colourless. Alkane: stays orange/purple
-OH (alcohol or acid)	PCl₅ (solid, dry sample)	steamy/misty fumes (HCl)	Not "white smoke" . Water also reacts, so the sample must be dry. Alternatives: Na → effervescence
1°/2° alcohol, aldehyde	$\text{K}_2\text{Cr}_2\text{O}_7$ + dilute H_2SO_4 , warm	orange → green	3° alcohol stays orange . Never HCl or HNO_3
Aldehyde	Fehling's/Benedict's, warm	blue solution → (brick-)red precipitate	Tollens': silver mirror . Ketone: no change
Carboxylic acid	$\text{Na}_2\text{CO}_3/\text{NaHCO}_3(\text{aq})$ or Mg	effervescence ; gas turns limewater cloudy	Blue litmus → red also accepted
$\text{CH}_3\text{CO}-$ / $\text{CH}_3\text{CH}(\text{OH})-$	I_2 + $\text{NaOH}(\text{aq})$	pale yellow ppt (antiseptic smell)	Distinguishes butanone from butanal
Halogenoalkane	Warm with ethanolic AgNO_3	white / cream / yellow ppt	Rate C-I > C-Br > C-Cl ; tertiary > primary; rate $\propto 1/\text{time}$
Unsaturation / high C:H	Burn	smoky, sooty yellow flame	

OXIDATION OF ALCOHOLS — THE APPARATUS LOGIC IS THE MARK

Want the aldehyde: heat with **distillation** apparatus so it **distils off as it forms**, preventing further oxidation to the acid.

Want the carboxylic acid or ketone: heat under reflux — vapour condenses and returns, so oxidation is complete. An open tube loses volatile (and flammable) reactants and products.

propan-1-ol → propanal → propanoic acid · propan-2-ol → propanone · 2-methylpropan-2-ol → **no reaction**.

$\text{C}_4\text{H}_{10}\text{O}$ primary alcohols butan-1-ol and 2-methylpropan-1-ol **both** give aldehydes, so the aldehyde test **cannot tell them apart**.

IR ranges: O-H alcohol **3750-3200** · O-H acid **3300-2500 (broad)** · C=O **1740-1680** · C=C 1669-1645 · C-H alkane 2962-2853 · C-Cl 700-550. **Following a reaction by IR:** the alcohol O-H peak (~3350) **decreases**, or the C-Cl peak (~590) **appears**.

Mass spectrum: M^+ at the highest $m/z = \text{M}_r$; **fragments must carry a + charge** (radicals are not detected); **isomers share M^+** . 15 CH_3^+ · **31 CH_2OH^+ (primary alcohol)** · 43 CH_3CO^+ · **45 CH_3CHOH^+ (secondary)**. Br compounds give **peaks in pairs** ($^{79}\text{Br}/^{81}\text{Br}$). 2-hydroxypropanal gives 15; 3-hydroxypropanal gives 31.

SHEET 3 — APPARATUS & THE "WHY" ANSWERS

Reflux — diagram mark points		Distillation — diagram mark points
Round-bottomed or pear-shaped flask (never conical) with liquid, and heat · vertical condenser with a water jacket, water in at the bottom, out at the top · open at the top , no gaps, no stopper · anti-bumping granules		Round flask + heat + anti-bumping granules · still head with the thermometer bulb level with the side arm (not in the liquid) · sloping Liebig condenser, water in at the lower end · sealed on the left, open receiver or vent on the right · drawn as separate pieces — a one-piece drawing loses a mark
Step	The answer that scores	
Anti-bumping granules	Provide sites/surfaces for small bubbles to form → smooth boiling. "Prevent bumping" alone is ignored	
Condenser water in at the bottom	The jacket fills completely with no air bubbles → more efficient cooling	
Electric heater / water bath, no Bunsen	Organic liquids are flammable	
Cool / add acid slowly while mixing	The reaction is exothermic ; a volatile reactant/product would evaporate or the mixture boil over	
Grind the solid	Larger surface area → faster reaction	
Wash with Na ₂ CO ₃ /NaHCO ₃	Neutralise remaining acid	
Open the tap while shaking	Release pressure (CO ₂ or vapour builds up)	
Why not NaOH to neutralise	It reacts with / hydrolyses the halogenoalkane — it is a nucleophile	
Wash with water	Removes water-soluble / ionic impurities	
Anhydrous CaCl ₂ / MgSO ₄ / Na ₂ SO ₄	Drying agent (never "dehydrating agent"); liquid cloudy → clear ; then decant or filter. Unsuitable: conc. H ₂ SO ₄ , CuSO ₄ , NaOH, CaCO ₃	
Not dry by heating (bromoethane)	The product is volatile and would evaporate first	
Collect the distillate over a range	Start just below and end just above the b.p. — cyclohexene (83 °C) 80–86; 1-bromobutane (102 °C) 100–104. Do not start or end on the b.p. itself	
Fractionating column instead of a still head	Better separation; acts like reflux so the higher-boiling reactant returns → higher yield	
Check purity	Measure the boiling temperature and compare with a data book , or use IR	
% yield below 100%	Incomplete reaction · side reactions · transfer losses · product lost in purification	
Remove the delivery tube before stopping heating	Prevent suck-back: the tube cools, pressure falls, cold water is drawn into the hot tube, which cracks	
Clean Mg ribbon with emery paper	Remove the MgO layer so only Mg is weighed	
Polystyrene cup inside a beaker	Stops the cup tipping over (stability) — not insulation	
Heat to constant mass	Weigh, heat, cool, reweigh; repeat until the mass is unchanged → reaction complete	

Separating funnel: tap with stem at the bottom, narrow stoppered neck, **two labelled layers**. Denser than water (lower layer): bromoethane, 1-bromobutane, 1-bromopropane. Less dense (upper): cyclohexene, 2-chloro-2-methylpropane, pentane, cyclohexane.

SHEET 4 — TITRATION

MAKING A STANDARD SOLUTION — 4-6 MARKS

A **standard solution** = a solution of **accurately known concentration**.

1. Calculate the mass: $n = c \times V$, $m = n \times M$ · **2.** weigh accurately; dissolve in a **small volume** of distilled water in a beaker, stirring · **3.** transfer through a funnel to a **volumetric flask, rinsing beaker and funnel and adding the washings** · **4.** make up to the mark, **dropwise at the end, bottom of the meniscus on the line, at eye level** · **5.** stopper and invert **several times**. **Never filter into the flask.**

Procedure: rinse the **burette and pipette with the solution they will hold** (water would dilute it); the **conical flask with distilled water only** — extra water does not change moles. **Volumetric pipette with a filler** for 25.0 cm³. Fill the burette **below eye level**. **Rough titration first** (finds the approximate end point, saving time), then add quickly to 1–2 cm³ below it and **dropwise with swirling**.

Record to 2 d.p. ending in 0 or 5 (24.20, not 24.2). **Concordant = within 0.10 cm³** (0.20 accepted). **Mean concordant titres only; exclude the rough.** If two already agree, a further titration is not needed. Dilute $\times 10$ first if the titre would otherwise exceed the burette.

Indicator	Titration	Colour change
Methyl orange	Acid added to alkali / carbonate	yellow → orange (never red)
Phenolphthalein	Alkali added to acid	colourless → (pale) pink
Phenolphthalein	Acid added to alkali	pink → colourless

Error	Effect on titre	Reason
Burette rinsed with water	Increases	Titrant diluted
Pipette rinsed with water	Decreases	Fewer moles in the flask
Conical flask wet	No change	Moles unchanged
Pipette blown out	Increases	More than 25.0 cm ³ delivered
NaOH titrant left open to air	Increases	Reacts with CO ₂ , so less NaOH
Water evaporates from the NaOH titrant	Decreases	More concentrated

THE CALCULATION CHAIN — THE SCALE-UP IS THE MOST-LOST MARK

mol titrant = (titre ÷ 1000) $\times c \rightarrow$ **equation ratio** $\rightarrow$ mol in the aliquot $\rightarrow \times$ **(250/25) for the whole flask** $\rightarrow$ mass ($\times M$) or concentration ($\div V$) $\rightarrow$ % or M_r or n .

Ratios: Na₂CO₃ + 2HCl $\rightarrow$ 1:2 · Ba(OH)₂, Ca(OH)₂, H₂SO₄, diprotic acids $\rightarrow$ 1:2 · citric acid $\rightarrow$ 1:3.

Back titration: mol acid added – mol acid left (from the titration, scaled) = mol that reacted.

Water of crystallisation: M_r of hydrate – M_r anhydrous = mass of water $\rightarrow \div 18 \rightarrow n$ (**must be a whole number**). **Identify a metal:** $A_r = M_r -$ the known non-metal part. **Never give a % above 100.**

SHEET 5 — ENTHALPY, UNCERTAINTY & GRAPHS

SOLUTION CALORIMETRY — THE METHOD AND WHY YOU EXTRAPOLATE

1. Known volume into a **polystyrene cup inside a beaker** (for **stability**) · **2.** record the temperature **every 30 s for ~3 min** to get a steady start · **3.** add the second reagent at a **known time**, stir, keep recording every 30 s — **do not record at the moment of mixing** · **4.** plot temperature against time, draw a best-fit line through the cooling points and **extrapolate back to the time of mixing**; ΔT = extrapolated value – starting value.

Why extrapolate: the reaction is **not instantaneous**, so **heat is lost while it happens**; extrapolation corrects for this.

Q = mc ΔT (m = **mass of solution** = total volume; c = 4.18) → **$\Delta H = -Q \div (1000 \times \text{mol of the limiting reactant})$** , in kJ mol⁻¹, **negative for a temperature rise**.

Assumptions (1 mark each): density = 1 g cm⁻³ · specific heat capacity = that of water · heat capacity of cup/thermometer negligible · mass of solid ignored.

Endothermic dissolving: extrapolate to the **minimum**. **Glass beaker instead of polystyrene:** value **less exothermic**, because glass conducts heat and has a higher heat capacity.

Spirit burner combustion: measure the mass/volume of water, initial and final water temperature, and the **mass of the burner before and after**. **Cap the burner when not lit and reweigh at once** (fuel evaporates).

Why much less exothermic than the data book: heat lost to surroundings and apparatus · **incomplete combustion (soot)** · fuel evaporates from the wick · non-standard conditions. **Burner lit before weighing** → mass burned overstated → value **less** exothermic.

Improvements: draught shield · lid on the water container · copper can · stir the water · shorter distance to the flame. **Not credited:** "repeat the experiment", "more accurate balance", "better insulation" with no detail.

UNCERTAINTY — THE ONE PLACE ROUNDING IS STRICTLY CHECKED

% uncertainty = (uncertainty per reading × number of readings ÷ measured value) × 100

Titre (two burette readings): $\pm 0.05 \times 2 = \pm 0.10 \text{ cm}^3$ → e.g. $0.10/11.70 \times 100 = 0.85\%$ · **temperature change** (two readings): $\pm 0.1 \times 2$ · **mass by difference** (two weighings): 2×0.0025 · **pipette or volumetric flask:** **one** reading.

Round correctly: 0.8547 → 0.85, not 0.86.

% error = |experimental – data book| ÷ the DATA BOOK value × 100 — divide by the data value, not yours.

Is the result consistent? Find the range: experimental $\pm$ (total % uncertainty × experimental value). **Inside → the difference is explained by measurement uncertainty; outside → there is a systematic error** (heat loss, gas escape).

Reduce % uncertainty: measure a **bigger quantity** — larger titre, larger ΔT , larger mass. **Uncertainty** gives the range the true value lies in; **accuracy** is how close you are to the true value. An **anomaly** need not be repeated if all the other points lie on the line.

Graphs: independent variable on **x**; labels **with units** ("Time / s"); **linear scales** — a non-uniform scale loses **both** plotting marks; points covering **at least half the grid**, plotted to within half a small square; best-fit straight line or smooth curve, **ignoring the anomaly**; **show construction lines on the graph**.

Rate at a point: tangent, gradient = $\Delta y/\Delta x$, **with units** (cm³ s⁻¹). **Rate from time:** rate $\propto 1/t$, units s⁻¹.

SHEET 6 — GASES, YIELDS & HAZARDS

Molar gas volume = $24.0 \text{ dm}^3 \text{ mol}^{-1}$ = $24\,000 \text{ cm}^3$ at r.t.p.

Identify a Group 2 carbonate: $\text{mol CO}_2 \rightarrow M_r \text{ of MCO}_3 = \text{mass} \div \text{mol} \rightarrow A_r = M_r - 60 \rightarrow$ nearest Group 2 element.

Measure the molar volume: $\text{mol metal} \rightarrow V_m = \text{volume} \div \text{mol}$, **units required.**

Why a measured V_m is too low: gas escapes before the bung is inserted · **CO₂ dissolves in water** · the metal is partly oxidised. Using **weak** ethanoic acid instead of HCl slows the reaction so **less gas escapes**.

Gas M_r from a lighter: $M_r = \text{mass lost} \div (V \div 24\,000)$. **Level the water inside and outside** the cylinder so the gas is at atmospheric pressure. If gas is lost, the mass loss seems smaller $\rightarrow M_r$ **too small**.

PERCENTAGE YIELD — THE ZERO-SCORING SHORTCUT

Liquid reactant: **mass = volume × density** $\rightarrow$ mol reactant $\rightarrow$ mol product (**equation ratio**) $\rightarrow$ theoretical mass $\rightarrow$ % **yield = actual $\div$ theoretical $\times 100$.**

Dividing masses directly (product g $\div$ reactant g) scores zero.

Empirical formula from combustion: weigh the **silica gel tube (H₂O)** and the **soda-lime tube (CO₂)** before and after. Use **pure O₂, not air** — air contains CO₂ and H₂O, and pure O₂ ensures complete combustion. **Mass O = sample – C – H.**

Precipitate-height graph: two straight lines; the **break point = the volume where the reactant stops being in excess**; the mole ratio there gives M_r .

Symbol	Meaning	Precaution that scores
Flame	Flammable	No naked flames ; electric heater / water bath
Flame over circle	Oxidising	Keep away from flammable materials
Skull and crossbones	Toxic	Fume cupboard (for Cl ₂ , Br ₂ , NO ₂ , SO ₂ , HCl)
Liquid dripping onto a hand	Corrosive	Gloves ; dilute; use a tray
Exclamation mark	Harmful / irritant	Gloves; avoid splashes into the eyes
Person with star on chest	Serious health hazard	Fume cupboard; avoid inhalation
Dead tree and fish	Harmful to the environment	Organic waste bottle, not down the sink

Precautions must match the hazard. Goggles, lab coat and "be careful" score nothing. Excess Cl₂ is absorbed in NaOH: $2\text{NaOH} + \text{Cl}_2 \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

FINAL CHECKLIST

Every observation: **colour + state**. Every test: **reagent + condition + result**. Colour changes: **both colours, in order**. Methyl orange $\rightarrow$ **orange**; phenolphthalein $\rightarrow$ **pink/colourless**. Sulfate test: **HCl + BaCl₂**. Halide test: **HNO₃ + AgNO₃**. Flame test: **nichrome, conc. HCl, non-luminous flame**. Titres to **2 d.p.**, mean of **concordant only**. $\times 10$ for 250 cm^3 from 25.0 cm^3 . $Q = mc\Delta T$ with the **mass of solution**. % uncertainty: **two readings $\rightarrow$ double**. % yield **via moles**. Diagrams: **no gaps, not sealed, correct water flow, flask not conical**. **Drying agent, not dehydrating**. **Specific hazard $\rightarrow$ specific precaution**. Never a % above 100 or an n that isn't a whole number.